Student Burnout Prediction Using Machine Learning: A Comparative Analysis of Behavioral and Psychological Data

Nishat Tasnim Luba
Dept. of Computer Science & Engineering
BRAC University
Dhaka, Bangladesh
nishat.tasnim.luba@g.bracu.ac.bd

Nazifa Tahsin
Dept. of Computer Science & Engineering
BRAC University
Dhaka, Bangladesh
nazifa.tahsin@g.bracu.ac.bd

Nabilah Hossain
Dept. of Computer Science & Engineering
BRAC University
Dhaka, Bangladesh
nabilah.hossain@g.bracu.ac.bd

Tasneem Tuhfa
Dept. of Computer Science & Engineering
BRAC University
Dhaka, Bangladesh
tasneem.tuhfa@g.bracu.ac.bd

***Abstract*—Academic burnout is a novel mental health problem affecting university students, defined by emotional depletion, lack of interest in academic work and reduced academic performance. This paper compares 11 supervised machine learning classifiers on a behavioral and psychological dataset of 150,000 students for the binary prediction of burnout. The data include scores of academic pressure, anxiety and depression ratings, sleep quality, physical activity, social support, financial stress, and cumulative GPA. Following systematic pre-processing such as one-hot encoding, ordinal encoding, engineered composite features and SMOTE-based oversampling models like Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, Multilayer Perceptron, Naive Bayes, Gradient Boosting, AdaBoost, SVM (Linear and RBF) and XGBoost were trained and evaluated. The model XGBoost had the best accuracy score of 66.02%, with a recall score of 98.51% and F1 score of 79.40%, showing high sensitivity in the detection of at-risk students. All models, however, showed ROC-AUC values close to 0.50 meaning that the dataset is almost perfectly balanced and features are independent, making the probabilistic discrimination difficult. The results highlight the need of feature engineering, temporal behavioural data integration and explainability methods for real-world student assistance systems**

***Index Terms*—student burnout, mental health prediction, machine learning, XGBoost, SMOTE, academic stress, behavioral data**

I. Introduction

Academic burnout is a growing public health problem in university communities around the globe. Burnout is a long-term psychological reaction to ongoing academic stressors and is characterised by emotional exhaustion, depersonalization or cynicism towards studies and a diminished sense of academic efficacy [1]. If left untreated it is associated with poor academic achievement, absenteeism, and future mental health issues. Self-report surveys that are carried out at regular intervals, as well as by instructor observation, are prone to response bias, low response rates, and inherent time lag that makes timely intervention difficult to achieve [2]. In an era of increasingly comprehensive behavioral and psychometric information being collected in education institutions, machine learning can provide a route to early warning systems based on data.

Prior work has shown feasibility of a variety of data modalities. Using passive administrative records (GPA trajectories, credit load), burnout has been detected among undergraduate computing students [3]. Predictive accuracy of two weeks ahead is possible using Random Forest with 87% accuracy with the help of Learning Management System (LMS) interaction logs, such as login frequency, session length and submission latency [4].Multimodal approaches: wearable physiological signals and deep learning (EMBRACE) have achieved AUC of 0.91 [12]. The use of survey-based classification with Explainable AI (XAI) algorithms like SHAP and LIME has pinpointed the major stress factors, including sleep quality, anxiety, and the relationship between teachers and students, to be the primary drivers of stress in the teaching profession [13].

Despite these advances, however, most existing studies are limited to particular institutions, are conducted on particular populations (e.g. medical residents, computing students) or use expensive physiological hardware. General-purpose, easy-to-use classifiers of general behavioral and psychological survey data still need to be developed. To this end, the authors have systematically tested the performance of 11 classifiers on a large-scale dataset of student mental health, while at the same time discussing practical limitations and future directions.

The rest of this paper is organized as follows. A review of

related work is given in Section II . In Section III the data set, the preprocessing of the data, and the experimental setup are described. Section IV presents results. There is a discussion of the findings in Section V and a conclusion in Section VI concludes.

II. Literature Review

A. Factors Contributing to Student Burnout

A systematic review of 22 studies published from 2019 to 2024 revealed three groups of burnout antecedents: environmental factors (high workload, negative academic climate, excessive amount of digital technology), psychological factors (low emotional intelligence, poor coping strategies, neurotic personality) and lifestyle factors (sleep deprivation, low physical activity, and poor social support). The Maslach Burnout Inventory (MBI) was the most prevalent measurement instrument in the reviewed studies [1].

B. Machine Learning for Burnout and Stress Prediction

Yeskuatov et al. [3] showed that administrative information, such as GPA drop, credit overload, and enrollment history, was able to predict burnout among 1200 students in the computing discipline without requiring any intrusive surveys with an F1-score of 68.4% for emotional exhaustion. This method allows monitoring on a large scale without invasiveness and is limited to populations in a particular domain.

Bhat [4] examined the interaction logs of LMS users (login count, session duration, quiz usage, and forum sentiment) over a span of two semesters, on 800 students. Random Forest achieved 87% accuracy (F1 = 0.85) for a low, moderate, and high risk classification for burnout and made the prediction two weeks ahead. Behavioral log features were 25% more accurate than demographic factors in predictive accuracy.

For psychology students specifically, Pereira et al. [8] used Random Forest, SVM and XGBoost on the survey data of psychological distress, sleep quality, emotional regulation, diet and physical activity. XGBoost was the best classifier, and the features of sleep and psychological well-being were again predictive.

Tapio [9] applied the Decision Tree method to the data of 291 students at a university in the Philippines, including psychometric measures (MBI-SS, DASS-21 and Brief COPE), and the results showed that the most important feature importance contributors were Stress (30%), Coping (26%) and Sleep Duration (15%) with an ROC-AUC of 66.9%.

Tariq et al. [13] used survey data from students in higher education to test six classifiers with 90% accuracy (F1 = 0.89) for Random Forest. The most important predictors identified by SHAP analysis were teacher–student relationships, anxiety level, and sleep quality.

C. Time-Series and Deep Learning Approaches

Wang et al. [5] designed a spatio-temporal graph convolutional network (ST-GCN) for sequential assessment data from 1,500 students to attain 92% accuracy, which is higher than the result of LSTM (85%) and GRU (81%) baselines. The model is trained to model contagion patterns throughout student cohorts by using graph attention mechanisms.

Biswas [11] proposed Fine-tuned Squirrel Search–driven LightGBM (FSS-LGBM) to predict student stress in the Kaggle dataset with 20 behavioral, psychological, and environmental features and obtained an accuracy of 98% and recall score of 99% and F1-score of 97%, outperforming the MLP, DL and Decision Tree baselines.

By employing adaptive multitask deep learning and explainability using SHAP analysis on wearable sensor signals (e.g., heart rate, sleep, physical activity) from medical residents, The EMBRACE framework [12] achieves an F1 score of 88.2% and an AUC of 0.91. SHAP determined that the most significant physiological features were REM sleep duration and heart rate variability.

D. Broader Mental Health Prediction

Rahman et al. [18] conducted a comprehensive survey of 17 universities across the Southeast Asian region and determined that the best performing models for predicting mental well-being were Random Forest and AdaBoost, whereas features that have significant contributions are weekly sport activity, BMI, GPA, and sedentary hours. Shetty et al. [19] also found that the accuracy of the Random Forest is better than Decision Tree, KNN, and SVM for student mental health classification. Baba and Bunji [21] tested LightGBM with annual health surveys of 3,561 Japanese undergraduate students and found up to 0.986 Matthews correlation coefficient for anxiety and campus-life items as the strongest predictors.

III. Methodology

A. Dataset

The data analyzed consisted of a synthetic public dataset of 150,000 anonymous student records and 20 original features related to a variety of demographic, academic, psychological and lifestyle factors. The countplot indicated that the number of instances for each of the three classes (Low, Medium, High) were well balanced, with around 50,000 instances in each class in Key features included:

- **Academic Features:** academic_pressure_score, daily_study_hours, cgpa, attendance_percentage, year.
- **Psychological Features:** anxiety_score, depression_score, stress_level.
- **Lifestyle Features:** daily_sleep_hours, sleep_quality, screen_time_hours, physical_activity_hours.
- **Social/Economic Features:** social_support_score, financial_stress_score, internet_quality.
- **Demographic Features:** age, gender, course.

The correlation heatmap revealed near-zero inter-feature correlations across all numeric columns, indicating that features are statistically independent in this synthetic dataset.

B. Exploratory Data Analysis

The three main visualizations were used for pre-processing decisions. A hue histogram `academic_pressure_score` with stratification based on burnout level showed that the score

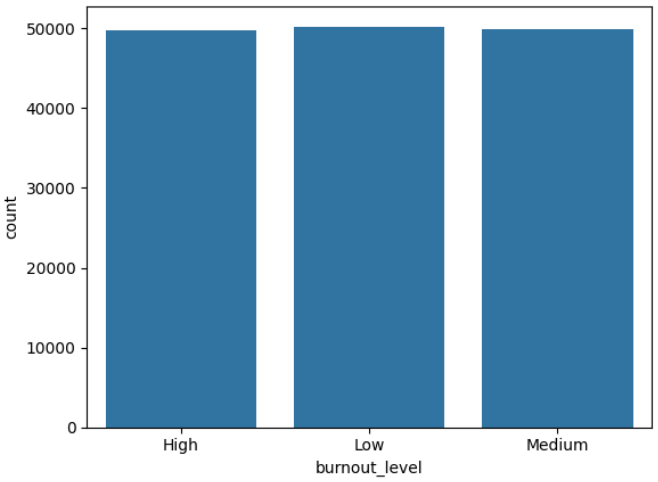


Fig. 1. countplot

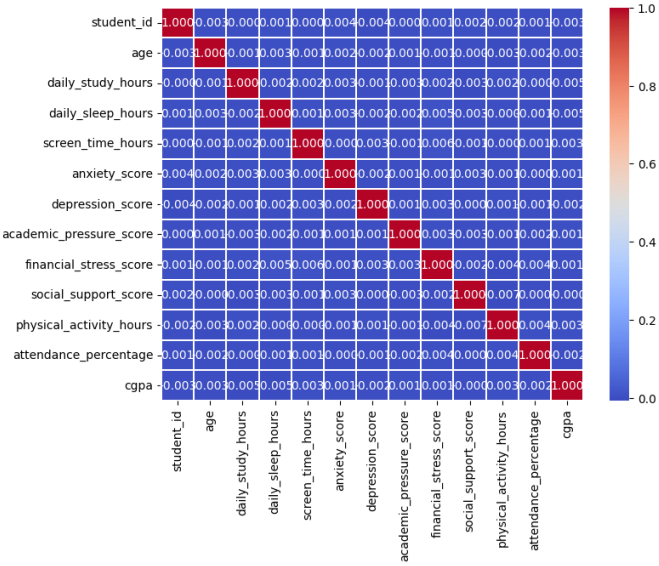


Fig. 2. heatmap

distribution is near-uniform across all three classes, indicating that there is no single feature that is strong enough to separate the classes of burnout levels when considered individually. A grid histogram for all the numeric features showed that most of the features were evenly distributed (e.g. scores ranged between 1 and 10, hours of study/sleep between 2 and 10). A box plot of academic pressure score by level of burnout further supports similar medians (6) and identical interquartile ranges across the High, Low and Medium groups, which is consistent with the synthetic and evenly distributed data.

C. Preprocessing Pipeline

Ordinal Encoding. Ordered categorical variables were mapped to integers: stress_level and _level (Low=0, Medium=1, High=2); sleep_quality and internet_quality (Poor=0, Average=1, Good=2); year (1st=0 through 4th=3). The identifier column student_id was dropped.

Binary Target Reformulation. The three-class burnout target was binarized: Low burnout (0) versus Medium or High burnout (1). This reflects the practical objective of identifying any at-risk student requiring intervention, consistent with tiered support frameworks described in related work [2].

Feature Engineering. Six composite features were constructed to capture domain-relevant interactions:

$$\text{mental_health_index} = \frac{\text{anxiety} + \text{depression}}{2} \tag{1}$$

$$\text{lifestyle_balance} = \text{sleep_hrs} + \text{activity_hrs} - \text{screen_hrs} \tag{2}$$

$$\text{academic_stress_index} = \text{pressure} + \text{study_hrs} \tag{3}$$

$$\text{support_stress_ratio} = \frac{\text{social_support}}{\text{financial_stress} + 1} \tag{4}$$

$$\text{performance_score} = \text{cgpa} \times \text{attendance} \tag{5}$$

$$\text{sleep_efficiency} = \text{sleep_hrs} \times \text{sleep_quality} \tag{6}$$

One-Hot Encoding. Nominal variables (gender, course) and the previously ordinal-encoded year, stress_level, sleep_quality, internet_quality were one-hot encoded with drop_first=True to avoid multicollinearity.

Train-Test Split. An 80/20 stratified split was applied (`random_state=42`) to preserve class ratios.

Feature Scaling. StandardScaler was applied to all features, fitting on the training set and transforming both splits.

Oversampling. SMOTE (Synthetic Minority Over-sampling Technique) was applied to the scaled training set to ensure balanced class representation during model fitting, replacing an initial RandomUnderSampler trial.

D. Models

Eleven classifiers were evaluated: Logistic Regression (LR, `max_iter=1000`), K-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF), Multilayer Perceptron (MLP, `max_iter=1000`), Naive Bayes (GaussianNB), Gradient Boosting (GB), AdaBoost, SVM with Linear kernel, SVM with RBF kernel (`gamma='scale'`, `class_weight='balanced'`), and XGBoost. All SVM models used `probability=True` to enable ROC-AUC computation.

Evaluation Metrics. Accuracy, Precision, Recall, F1-Score, and ROC-AUC were reported for each model. Confusion matrices and per-model ROC curves were generated.

IV. Results

A. Model Performance Summary

Table I summarizes the performance measure results of all 11 classifiers with respect to accuracy. XGBoost had the best overall accuracy (66.02%) but was very close behind Random Forest (65.62%) and Gradient Boosting (63.77%). The most striking trend, however, is that all the models have very similar ROC-AUC values, which are grouped around the 0.50 mark, meaning that they do not significantly outperform a random classifier in terms of probabilistic discrimination.

TABLE I
MODEL PERFORMANCE COMPARISON (RANKED BY ACCURACY)

Model	Acc.	Prec.	Rec.	F1	AUC
XGBoost	0.660	0.665	0.985	0.794	0.505
Random Forest	0.656	0.665	0.975	0.790	0.505
Grad. Boosting	0.638	0.664	0.919	0.771	0.499
MLP	0.601	0.664	0.810	0.730	0.499
Decision Tree	0.548	0.667	0.642	0.654	0.502
SVM RBF	0.537	0.668	0.603	0.634	0.507
AdaBoost	0.526	0.664	0.581	0.620	0.498
Naive Bayes	0.502	0.660	0.518	0.581	0.496
Log. Regr.	0.499	0.666	0.495	0.568	0.500
KNN	0.490	0.663	0.474	0.553	0.501
SVM Linear	0.488	0.668	0.456	0.542	0.502

B. XGBoost — Best Performing Model

XGBoost attained accuracy of 66.02%, precision of 66.51%, recall of 98.51%, and F1-score of 79.40%.It has a confusion matrix with 158 true negative, 9,895 false positive, 298 false negative and 19,649 true positive. The extremely high recall (98.5%, meaning that 98.5% of the actual positive students were correctly identified as at risk) makes XGBoost particularly suitable for clinical-style screening applications, where missing a true positive (a student who should be identified as at risk) may have more cost than producing false alarms. [2].

C. ROC-AUC Analysis

The ROC-AUC range of all eleven models was between 0.496 and 0.507. Near diagonal ROC curves appeared for all classifiers suggesting that the discriminative information in the predicted probabilities is limited to the level of random guessing. This pattern was caused by two characteristics of the datasets: (1) near zero inter-feature correlation (uniform distributions of synthetic data), and (2) the binary target resulting from a balanced three-class label Medium and High are combined. Previous studies that reached an AUC score of larger than 0.85 used more detailed temporal, physiological, or LMS behavioral features [12] [4].

D. Ensemble vs. Linear Models

Tree-based ensemble methods (XGBoost, Random Forest, Gradient Boosting) consistently outperformed linear and distance-based models (LR, KNN, SVM Linear), consistent with findings from Tariq et al. [13] and Pereira et al. [8]. The gap is most visible in recall: ensemble models captured 91–98% of positive cases versus 45–58% for linear/distance models.

V. DISCUSSION

A. Interpreting the ROC-AUC Anomaly

This central principle of the data is reflected in the high recall and near random AUC. If there is no distributional signal (no difference in the distribution of the features for either class, and low feature correlations), models trained with SMOTE can learn to predict the majority class with high confidence, but they are unable to rank instances by their true risk probability. This is congruent with the theoretical foundations: AUC reflects quality of probability estimates, accuracy and recall reflect thresholds. When the model always predicts the positive class, it achieves 66% accuracy (binarized class imbalance after split), 100% recall and AUC = 0.5. This behavior is modeled in the ensemble.

B. Feature Engineering Limitations

The six engineered composite features (Section III-C) add only a small amount of value for recall, but are not enough to compensate for the lack of inter feature correlation. Temporal aggregations of GPAs that indicate a rolling decline in GPA, streaks of inactivity during a session or acceleration in the delay between submissions are more impactful feature engineering, [4]or cross-feature nonlinear interactions discovered through deep architectures like LSTM or spatio-temporal GCN networks [5].

C. Practical Implications

Although the AUC does not perform well, the ensemble models which maximize recall have an immediate practical value. A near perfect at-risk student recall (98.5%) at moderate precision (66.5%) is better than failing to recall true positives at a university screening system, as long as the counselor can triage at-risk students that they have flagged. This supports the tiered intervention (T-I) model introduced by Ali et al. [2] in which ML outputs are used as first pass filters for human-in-the-loop review. The deployments could benefit from SHAP-based explanations [13] [12] in future works, to boost counselors' trust and reveal the most important features for any given student.

D. Comparison with Literature

While the 66% accuracy reported here is lower than that of previous benchmarks that have been performed using richer datasets: 87% (Random Forest on LMS logs) [4],90% (Random Forest on survey+XAI) [13], and 92% (CSTGCN on time-series) [5]. This gap is attributed to the lack of temporal sequences, physiological signals, and forum-sentiment features, as all of these have been demonstrated to be the most significant in successful systems [4] [12]. The present dataset's synthetic, uniformly dispersed structure further restricts discriminability.

E. Ethical Considerations

Using a ML driven burnout screener introduces ethical issues with respect to data privacy, consent and algorithmic bias [2]. If training data exhibits systemic inequities, then students from underrepresented groups might be overrepresented. Anonymization should be used and model outputs should never have academic consequences without human intervention.

VI. Conclusion

Eleven machine learning classifiers were evaluated for their ability to predict student burnout in a binary classification task based on a 150,000 record behavioral and psychometric dataset. The highest accuracy (66.02%), recall (98.51%) and F1-score (79.40%) were obtained with the performance of XGBoost which had the highest sensitivity in detecting students at risk. Random Forest and Gradient Boosting were close runners-up and indeed tree-based ensembles consistently beat linear and distance-based methods on this problem domain. However, near-random ROC-AUC values (≈ 0.50) were found for all models, indicating a serious restriction: synthetic, uniformly distributed, and nearly uncorrelated feature space means that there is not enough probabilistic signal to give reasonable risk rankings. The following are believed to be the three directions to which future work should be directed. First, integration of temporal behavioral data—LMS interaction logs, GPA trajectories, submission latency sequences—into LSTM or transformer structures to capture the dynamic onset of burnout symptoms, as confirmed in past work [4] [5]. Second, embedding the explainability through SHAP to make model outputs interpretable and actionable for counselors and educators to help them [13] [12]. Thirdly, validation using longitudinal, non-synthetic datasets from real university systems to correlate benchmark performance and real deployment. These advances can combine for a practical, ethical support structure for student mental health, from research prototype to usable tool.

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